

Bus Reservation And Ticketing System

<https://github.com/aiyshwary/Bus-Reservation-and-Ticketing-System>

OVERVIEW

A Java-based console application that simulates a complete bus ticket reservation and billing workflow, including seat management, and discount-based billing operations.

IMPACT

Built a fully functional console-based Bus Reservation and Ticketing System in Java with structured input validation, seat management, and automated billing with passenger-type discounts.

KEY DETAILS

1. Implements user authentication, seat selection, and multi-passenger booking in a single session.
2. Handles billing logic including discount tiers for different passenger categories.
3. Uses structured in-memory data storage for transaction tracking without a database dependency.
4. Validates all inputs to prevent double-booking and illegal seat selections.
5. Supports 5 Mindanao route destinations with preset fares and 20 seats each; applies a flat 20% discount for Citizen passengers.

6. Offers a 5-option console menu: Destination (view routes and live seat counts), Passengers (book), View (search and receipt), and Exit.

7. Enforces a 3-attempt login lockout; uses three parallel 2D arrays to store passenger name/destination